Mathematical Analysis 2

Multivariable derivatives

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Throughout, vectors are columns and all norms are Euclidean. Problems marked $\star$ are further practice.

Suggested route for a 90-minute tutorial: Problems 1(a,b), 2, 3(a,b), 4(a), 5, 6(a,c), and 7(a). Use the remaining parts for discussion or independent practice.

The derivative is a linear map

Let $U \subseteq \mathbb{R}^n$ be open, $F : U \to \mathbb{R}^m$, and $a \in U$. We say that F is **differentiable at** a if there is a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{h \to 0} \frac{\|F(a+h) - F(a) - L(h)\|}{\|h\|} = 0.$$

We write $DF(a) = L$. The derivative is a map acting on an increment h; $F(a)$ is a value.

1 Constant, linear, and affine maps

Use the definition above, rather than coordinate differentiation.

(a) Find $DF(a)h$ and the remainder $F(a+h) - F(a) - DF(a)h$ for

$$F(x) = c, \qquad F(x) = Ax, \qquad F(x) = Ax + b,$$

where $A \in \mathbb{R}^{m \times n}$ and $b, c \in \mathbb{R}^m$ are fixed. Give the matrix of $DF(a)$ in the standard bases and its size. Does it depend on a?

(b) Take

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \quad a = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad h = \begin{pmatrix} 0.02 \\ -0.01 \\ 0.03 \end{pmatrix}.$$

Compute $F(a)$, $DF(a)h$, and $F(a+h)$. Is the linear approximation exact?

(c) $\star$ Prove that the linear map in the definition is unique. *Hint: compare two candidates on increments $h = tv$.*

Solution

(a) For $F(x) = c$, take $DF(a)h = 0$. For $F(x) = Ax$ and for $F(x) = Ax + b$, take $DF(a)h = Ah$. In each case the remainder is identically zero, so the defining limit is zero. The matrix of $DF(a)$ is respectively the zero $m \times n$ matrix or A, independent of a. In particular, the constant b disappears on taking the difference $F(a+h) - F(a)$.

(b) Direct calculation gives

$$F(a) = \begin{pmatrix} 3 \\ -6 \end{pmatrix}, \qquad DF(a)h = Ah = \begin{pmatrix} -0.03 \\ 0.09 \end{pmatrix}, \qquad F(a+h) = \begin{pmatrix} 2.97 \\ -5.91 \end{pmatrix}.$$

The approximation is exact for every increment h because F is affine.

(c) Suppose L and M both satisfy the definition. Subtracting their remainder expressions and using the triangle inequality gives

$$\frac{\|(L-M)h\|}{\|h\|} \longrightarrow 0.$$

For a fixed $v \neq 0$ and $h = tv$, the quotient equals $\|(L-M)v\| / \|v\|$, independently of $t \neq 0$. It must therefore be zero. Thus $Lv = Mv$ for every v, including $v = 0$, so $L = M$.

2 Quadratic forms from the definition

Let $B \in \mathbb{R}^{n\times n}$ be fixed and $Q(x) = x^{\mathsf{T}}Bx$.

(a) Expand $Q(a+h) - Q(a)$. Identify the part linear in h and prove that the remaining term, divided by $\|h\|$, tends to zero. Give $DQ(a)h$ and the matrix of $DQ(a)$ in the standard bases, without assuming that B is symmetric.

(b) Simplify your answer when $B = B^{\mathsf{T}}$. Apply it to

$$Q(x,y) = 2x^2 - 2xy + 3y^2, \qquad a = (1,-1).$$

Use the derivative to approximate $Q(1.02, -0.99)$. Compute the exact remainder and compare with the exact value.

Solution

(a) Expanding gives

$$Q(a+h) - Q(a) = a^{\mathsf{T}}Bh + h^{\mathsf{T}}Ba + h^{\mathsf{T}}Bh.$$

The first two terms define a linear function of h, and

$$DQ(a)h = a^{\mathsf{T}}(B + B^{\mathsf{T}})h.$$

Its matrix in the standard bases is the $1 \times n$ row $a^{\mathsf{T}}(B + B^{\mathsf{T}})$. For example, with $C = \sum_{i,j} |B_{ij}|$, we have

$$|h^{\mathsf{T}}Bh| \leq C\,\|h\|^2, \qquad \frac{|h^{\mathsf{T}}Bh|}{\|h\|} \leq C\,\|h\| \longrightarrow 0.$$

This proves differentiability, not just the existence of partial derivatives.

(b) If $B = B^{\mathsf{T}}$, the derivative simplifies to $2a^{\mathsf{T}}Bh$. For the given form,

$$B = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}, \quad Q(1,-1) = 7, \quad DQ(1,-1)h = 6h_1 - 8h_2.$$

At $h = (0.02, 0.01)$ the predicted change is $0.12 - 0.08 = 0.04$. The exact remainder is

$$2(0.02)^2 - 2(0.02)(0.01) + 3(0.01)^2 = 0.0007.$$

Hence the linear approximation is 7.04 and the exact value is 7.0407.

Partial derivatives and directions

Derivatives along a line

For $F : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ and $v \in \mathbb{R}^n \setminus \{0\}$, define

$$D_v F(a) = \lim_{t \to 0} \frac{F(a+tv) - F(a)}{t}, \qquad \frac{\partial F}{\partial x_i}(a) = D_{e_i} F(a),$$

whenever the limits exist. Limits are **two-sided**. We do **not** automatically normalize v. For a scalar-valued function, the **gradient** is the column vector $\nabla f(a) = (\frac{\partial f}{\partial x_1}(a), \ldots, \frac{\partial f}{\partial x_n}(a))^{\mathsf{T}}$.

3 From the total derivative to partial derivatives

Suppose that $F : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at a.

(a) Prove directly from the definition that

$$D_v F(a) = DF(a)v.$$

(b) For $F = (f_1, \ldots, f_m)^{\mathsf{T}}$, the **Jacobian matrix** $J_F(a)$ is the $m \times n$ matrix with (j, i) entry $\frac{\partial f_j}{\partial x_i}(a)$. Prove that it represents $DF(a)$ in the standard bases. Deduce, for scalar-valued f, that $D_v f(a) = \nabla f(a)^{\mathsf{T}} v$.

(c) Find $\nabla Q(1, -1)$ for the quadratic form in Problem 2, and its directional derivative in the unit direction $v = (3/5, 4/5)$.

Solution

(a) Fix $v \neq 0$. Since $DF(a)$ is linear, $DF(a)(tv) = t\, DF(a)v$. Subtracting and adding this linear term gives

$$\begin{aligned} D_v F(a) &= \lim_{t \to 0} \frac{F(a+tv) - F(a)}{t} \\ &= \lim_{t \to 0} \left(\frac{F(a+tv) - F(a) - DF(a)(tv)}{\|tv\|} \frac{\|tv\|}{t} + DF(a)v \right). \end{aligned}$$

The scalar factor $\|tv\| / t$ is bounded, because for $t \neq 0$,

$$\left| \frac{\|tv\|}{t} \right| = \|v\|.$$

Differentiability at a, applied to $h = tv \to 0$, gives

$$\lim_{t \to 0} \frac{\|F(a+tv) - F(a) - DF(a)(tv)\|}{\|tv\|} = 0.$$

Thus the vector in the first fraction tends to zero, and multiplying it by the bounded scalar factor still gives a vector tending to zero. Consequently the directional derivative exists and

$$D_v F(a) = 0 + DF(a)v = DF(a)v.$$

(b) Put $v = e_i$. Then $\frac{\partial F}{\partial x_i}(a) = DF(a)e_i$, the ith column of the matrix representing $DF(a)$. Its jth component is $\frac{\partial f_j}{\partial x_i}(a)$, exactly the (j, i) entry of $J_F(a)$. Thus the Jacobian represents the total derivative: $DF(a)h = J_F(a)h$. For scalar f, the Jacobian is a row, and

$$J_f(a) = \nabla f(a)^{\mathsf{T}}, \qquad D_v f(a) = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(a) v_i = \nabla f(a)^{\mathsf{T}} v.$$

(c) Here $\nabla Q(1, -1) = (6, -8)^{\mathsf{T}}$, so $D_v Q(1, -1) = 6 \cdot 3/5 - 8 \cdot 4/5 = -14/5$.

4 Computing partial derivatives

For each function, find its open domain, compute all first partial derivatives, and evaluate the gradient at the indicated point. Explain why the function is differentiable there. Then compute $D_v f(a)$.

(a) $f(x,y) = e^{xy}, \quad a = (1,0), \quad v = (1,-1).$

(b) $f(x,y) = \ln(x^2 - 2x - y), \quad a = (0,-1), \quad v = (1,2).$

(c) $f(x,y,z) = (x/y)^z, \quad a = (e,1,2), \quad v = (0,1,-1).$ Interpret a variable real power by $u^z = \exp(z \ln u)$ for $u > 0$.

Solution

All three functions are compositions of smooth functions on their stated open domains. Thus they are differentiable there and the gradient formula for directional derivatives applies.

(a) The domain is $\mathbb{R}^2$, and

$$\frac{\partial f}{\partial x} = ye^{xy}, \qquad \frac{\partial f}{\partial y} = xe^{xy}, \qquad \nabla f(1,0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Hence $D_{(1,-1)} f(1,0) = -1$.

(b) The domain is the open set $\{(x,y) : x^2 - 2x - y > 0\} = \{(x,y) : y < x^2 - 2x\}$. Writing $u = x^2 - 2x - y$, we obtain

$$\frac{\partial f}{\partial x} = \frac{2x-2}{u}, \qquad \frac{\partial f}{\partial y} = -\frac{1}{u}, \qquad \nabla f(0,-1) = \begin{pmatrix} -2 \\ -1 \end{pmatrix}.$$

Thus $D_{(1,2)} f(0,-1) = -2 - 2 = -4$.

(c) Under the stated real-power convention the open domain is

$$\{(x,y,z) : y \neq 0,\ x/y > 0\} = \{(x,y,z) : xy > 0\}.$$

Using $f = \exp(z \ln(x/y))$ gives

$$\frac{\partial f}{\partial x} = f\frac{z}{x}, \qquad \frac{\partial f}{\partial y} = -f\frac{z}{y}, \qquad \frac{\partial f}{\partial z} = f \ln(x/y).$$

At $(e,1,2)$, the function value is e^2, so

$$\nabla f(e,1,2) = \begin{pmatrix} 2e \\ -2e^2 \\ e^2 \end{pmatrix}, \qquad D_{(0,1,-1)} f(e,1,2) = -3e^2.$$

The prescribed vector has not been normalized; replacing it by a unit vector would change the answer by a factor $1/\sqrt{2}$.

5 Why the gradient gives steepest ascent and descent

Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ be differentiable at a.

(a) Assume $\nabla f(a) \neq 0$. Prove that among *unit* vectors, the unique direction of steepest ascent is $\nabla f(a)/\left\|\nabla f(a)\right\|$, and the unique direction of steepest descent is its negative. Find the corresponding rates of change. *Hint: use Cauchy–Schwarz and its equality condition.*

(b) What happens if $\nabla f(a) = 0$? Does this imply that a is a local maximum or minimum? Give a counterexample if not.

(c) Apply the result to $g(x, y) = x^2 + 4y^2$ at $(1, 1)$. Also compute the derivatives for $v = (1, 0), (0, 1), (1, 1)$ and $(1, 1)/\sqrt{2}$. Why must directions have equal length when comparing rates of change per unit distance?

Solution

(a) Since f is differentiable at a, Problem 3 gives

$$D_v f(a) = Df(a)v = \nabla f(a)^{\mathsf{T}} v = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(a) v_i.$$

To compare rates per unit distance, restrict to $\|v\| = 1$. By Cauchy–Schwarz,

$$\begin{aligned} |D_v f(a)| &\leq \left(\sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}(a) \right)^2 \right)^{1/2} \left(\sum_{i=1}^{n} v_i^2 \right)^{1/2} \\ &= \left\|\nabla f(a)\right\| \|v\| = \left\|\nabla f(a)\right\|. \end{aligned}$$

Thus every unit direction satisfies

$$-\left\|\nabla f(a)\right\| \leq D_v f(a) \leq \left\|\nabla f(a)\right\|.$$

An upper bound alone does not show that it is attained. Since $\nabla f(a) \neq 0$, define

$$u = \frac{\nabla f(a)}{\left\|\nabla f(a)\right\|}.$$

This is a unit vector. Substituting its coordinates into the directional derivative gives

$$\begin{aligned} D_u f(a) &= \frac{1}{\left\|\nabla f(a)\right\|} \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}(a) \right)^2 \\ &= \frac{\left\|\nabla f(a)\right\|^2}{\left\|\nabla f(a)\right\|} = \left\|\nabla f(a)\right\|. \end{aligned}$$

Hence u attains the largest possible rate: it is the direction of steepest ascent. The vector $-u$ is also a unit vector, and linearity gives

$$D_{-u} f(a) = -D_u f(a) = -\left\|\nabla f(a)\right\|.$$

It therefore attains the smallest possible rate and gives steepest descent.

Finally, equality in Cauchy–Schwarz requires v to be parallel to $\nabla f(a)$. Among unit vectors, the positive parallel direction is exactly u, and the negative one is exactly $-u$. Thus both optimizing unit directions are unique.

Alternative proof using the angle. Assume $\nabla f(a) \neq 0$, put $u = \nabla f(a)/\|\nabla f(a)\|$, and let v be any unit vector. The angle $\theta \in [0, \pi]$ between $\nabla f(a)$ and v satisfies $\cos\theta = u^{\mathsf{T}} v$. The Euclidean dot-product identity and Problem 3 give

$$D_v f(a) = \nabla f(a)^{\mathsf{T}} v = \|\nabla f(a)\| \, \|v\| \cos\theta = \|\nabla f(a)\| \cos\theta.$$

Since $-1 \leq \cos\theta \leq 1$, the largest and smallest possible rates are $\|\nabla f(a)\|$ and $-\|\nabla f(a)\|$. They are attained at $v = u$ and $v = -u$, respectively. For uniqueness, note that

$$\|v - u\|^2 = 2 - 2\cos\theta, \qquad \|v + u\|^2 = 2 + 2\cos\theta.$$

Thus $\cos\theta = 1$ holds exactly when $v = u$, and $\cos\theta = -1$ exactly when $v = -u$. If the gradient is zero, the angle is undefined; part (b) below handles that case directly. This proof uses the standard Euclidean notion of angle, whose dot-product formula is justified by Cauchy–Schwarz.

Meaning of steepest. For a fixed unit vector v and $t \downarrow 0$, differentiability gives

$$f(a + tv) - f(a) = t\, D_v f(a) + o(t).$$

The gradient direction maximizes the coefficient of t, the rate of change per unit distance at a. It need not give the largest actual increase for a fixed positive step length, even when all steps have the same length. For example, $f(x, y) = x + 100y^2$ has gradient $(1, 0)$ at $(0, 0)$, but steps of length 0.1 give

$$f(0.1, 0) - f(0, 0) = 0.1, \qquad f(0, 0.1) - f(0, 0) = 1.$$

The derivative describes the first-order change as the step tends to zero.

(b) If $\nabla f(a) = 0$, every directional derivative is zero. Every unit vector attains both the largest and smallest first-order rate, which are both zero. There is no distinguished first-order ascent direction. This does not imply an extremum: $f(x) = x_1^3$ on $\mathbb{R}^n$ has gradient zero at the origin but takes both positive and negative values arbitrarily close to it.

(c) For $g(x, y) = x^2 + 4y^2$,

$$\nabla g(1, 1) = \begin{pmatrix} 2 \\ 8 \end{pmatrix}, \qquad \|\nabla g(1, 1)\| = \sqrt{68} = 2\sqrt{17}.$$

The unit directions of steepest ascent and descent are respectively $\pm(1, 4)/\sqrt{17}$, with rates $\pm 2\sqrt{17}$. The requested derivatives are

v	$(1,0)$	$(0,1)$	$(1,1)$	$(1,1)/\sqrt{2}$
$D_v g(1,1)$	2	8	10	$5\sqrt{2}$

For a direct check, along a general direction (p, q),

$$g(1 + tp, 1 + tq) = 5 + t(2p + 8q) + t^2(p^2 + 4q^2).$$

The derivative at $t = 0$ is therefore $2p + 8q$. The vector $(1, 1)$ has length $\sqrt{2}$, so the path with this velocity travels $\sqrt{2}$ times as fast as a path with a unit velocity. After normalization its rate is $5\sqrt{2} < 8$: per unit distance, the direction $(0, 1)$ gives faster increase than the diagonal. Without a length constraint, the directional derivative is unbounded above and below when the gradient is nonzero, simply by scaling the velocity.

Norms, composition, and linear approximation

6 The squared norm, the norm, and its reciprocal

Consider $S, N : \mathbb{R}^n \to \mathbb{R}$, where $n \geq 1$, given by

$$S(x) = \|x\|^2, \qquad N(x) = \|x\|.$$

(a) Use the definition to find $DS(a)h$ at every $a \in \mathbb{R}^n$.

(b) For $a \neq 0$, use the definition to prove that

$$DN(a)h = \frac{\langle a, h\rangle}{\|a\|}.$$

Give the gradient and bound the remainder by a constant times $\|h\|^2$ near $h = 0$. *Hint: rationalize $\|a+h\| - \|a\|$ and use the reverse triangle inequality.*

(c) Is N differentiable at 0? Does $D_v N(0)$ exist for a nonzero v under our two-sided convention? Compare with S.

(d) Find the derivative and gradient of $R(x) = 1/\|x\|$ on $\mathbb{R}^n \setminus \{0\}$.

Solution

(a) The identity

$$\|a+h\|^2 - \|a\|^2 = 2\langle a, h\rangle + \|h\|^2$$

shows that $DS(a)h = 2\langle a, h\rangle$, with gradient $2a$. The remainder divided by $\|h\|$ equals $\|h\| \to 0$. This works at every a, including $DS(0) = 0$.

(b) Fix $a \neq 0$, and set $r = \|a\|$, $s = \|a+h\|$ and $\delta = s - r$. Rationalizing gives

$$\delta = \frac{2\langle a, h\rangle + \|h\|^2}{s+r}.$$

Subtract the candidate linear term:

$$\delta - \frac{\langle a, h\rangle}{r} = \frac{r\|h\|^2 - \langle a, h\rangle\ \delta}{r(s+r)}.$$

The reverse triangle inequality gives $|\delta| \leq \|h\|$, and $|\langle a, h\rangle| \leq r\|h\|$. Since $s + r \geq r$,

$$\left| \|a+h\| - \|a\| - \frac{\langle a, h\rangle}{\|a\|} \right| \leq \frac{2\|h\|^2}{s+r} \leq \frac{2}{\|a\|}\|h\|^2.$$

Dividing by $\|h\|$ proves differentiability. Thus

$$DN(a)h = \frac{\langle a, h\rangle}{\|a\|}, \qquad \nabla N(a) = \frac{a}{\|a\|} \quad (a \neq 0).$$

(c) At the origin and for $v \neq 0$,

$$\frac{N(tv) - N(0)}{t} = \frac{|t|}{t}\|v\|.$$

The right and left limits are $\|v\|$ and $-\|v\|$. Thus no such two-sided directional derivative exists, and N cannot be differentiable at the origin. A one-sided definition with $t \downarrow 0$ would instead give $\|v\|$, which is not linear in v. In contrast, S is differentiable at the origin and $D_v S(0) = 0$.

(d) On $\mathbb{R}^n \setminus \{0\}$, compose N with $s \mapsto s^{-1}$:

$$DR(a)h = -\frac{\langle a, h\rangle}{\|a\|^3}, \qquad \nabla R(a) = -\frac{a}{\|a\|^3}.$$

7 Using the chain rule

You may use the chain rule

$$D(G \circ F)(a) = DG(F(a)) \circ DF(a), \qquad J_{G\circ F}(a) = J_G(F(a))J_F(a).$$

State the input and output dimensions of the maps you compose.

(a) **A first composition.** Let $F : \mathbb{R}^2 \to \mathbb{R}^2$ and $G : \mathbb{R}^2 \to \mathbb{R}$ be given by

$$F(x, y) = \begin{pmatrix} x + y \\ xy \end{pmatrix}, \qquad G(u, v) = u^2 + v.$$

Write down $H = G \circ F$. Compute $J_F(x, y)$ and $J_G(u, v)$, and use their product to find $J_H(x, y)$ and $J_H(1, 2)$. Check your answer by differentiating the explicit formula for H.

(b) **A tangent plane.** Let $f(x, y) = \sqrt{x^2 + 4y^2 + 1}$. Find $Df(1, 1)$ and the tangent plane to its graph at $(1, 1, f(1, 1))$. Use it to approximate $f(1.02, 0.99)$.

(c) **Least squares.** Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$. For

$$L(x) = \tfrac{1}{2} \|Ax - b\|^2 ,$$

find $DL(a)h$ and $\nabla L(a)$. Write the equation $\nabla L(a) = 0$ in matrix form.

(d) $\star$ **Normalizing a vector.** Let $T : \mathbb{R}^n \setminus \{0\} \to \mathbb{R}^n$, $T(x) = x/\|x\|$. Find $DT(a)h$ and its Jacobian. Show that the radial direction a lies in the kernel and that $DT(a)h$ is perpendicular to a.

Solution

(a) First write the composition explicitly:

$$H(x, y) = G(F(x, y)) = (x + y)^2 + xy = x^2 + 3xy + y^2.$$

The Jacobians of the two maps are

$$J_F(x, y) = \begin{pmatrix} 1 & 1 \\ y & x \end{pmatrix}, \qquad J_G(u, v) = \begin{pmatrix} 2u & 1 \end{pmatrix}.$$

The outer derivative must be evaluated at $F(x, y) = (x + y, xy)$. Hence

$$\begin{aligned} J_H(x, y) &= J_G(F(x, y))J_F(x, y) \\ &= \begin{pmatrix} 2(x + y) & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ y & x \end{pmatrix} \\ &= \begin{pmatrix} 2x + 3y & 3x + 2y \end{pmatrix}. \end{aligned}$$

The product has dimensions $(1 \times 2)(2 \times 2) = 1 \times 2$. At $(1, 2)$, this gives $J_H(1, 2) = (8, 7)$, or $DH(1, 2)h = 8h_1 + 7h_2$. Differentiating $H = x^2 + 3xy + y^2$ directly gives the same row.

(b) Write $f = G \circ F$, where $F : \mathbb{R}^2 \to (0,\infty)$, $F(x,y) = x^2 + 4y^2 + 1$, and $G : (0,\infty) \to \mathbb{R}$, $G(s) = \sqrt{s}$. The chain rule gives

$$Df(x,y)h = \frac{xh_1 + 4yh_2}{\sqrt{x^2+4y^2+1}}, \qquad Df(1,1)h = \frac{h_1 + 4h_2}{\sqrt{6}}.$$

The tangent plane is

$$z = \sqrt{6} + \frac{x - 1 + 4(y-1)}{\sqrt{6}}.$$

For $h = (0.02, -0.01)$ this gives

$$f(1.02, 0.99) \approx \sqrt{6} - \frac{0.02}{\sqrt{6}} \approx 2.441325.$$

The exact value is $\sqrt{5.9608} \approx 2.441475$.

(c) Compose $F : \mathbb{R}^n \to \mathbb{R}^m$, $F(x) = Ax - b$, with $G : \mathbb{R}^m \to \mathbb{R}$, $G(y) = \frac{1}{2}\|y\|^2$. By Problems 1 and 6, $DF(a)h = Ah$ and $DG(y)k = y^{\mathsf{T}}k$. Therefore

$$DL(a)h = (Aa - b)^{\mathsf{T}}Ah, \qquad \boxed{\nabla L(a) = A^{\mathsf{T}}(Aa - b)}.$$

The equation $\nabla L(a) = 0$ is the system of *normal equations* $A^{\mathsf{T}}A\,a = A^{\mathsf{T}}b$. No invertibility or rank assumption is needed for this identity.

(d) The scalar map $R(x) = 1/\|x\|$ takes $\mathbb{R}^n \setminus \{0\}$ into $\mathbb{R}$, and $T(x) = R(x)x$ takes it into $\mathbb{R}^n$. Using the product rule and Problem 6(d),

$$DT(a)h = \frac{h}{\|a\|} - \frac{a\,\langle a, h\rangle}{\|a\|^3}.$$

With $u = a/\|a\|$, this becomes

$$\boxed{J_T(a) = \frac{1}{\|a\|}(I - uu^{\mathsf{T}})}.$$

Substituting $h = a$ gives zero. Also

$$a^{\mathsf{T}}DT(a)h = \frac{\langle a, h\rangle}{\|a\|} - \frac{\|a\|^2\,\langle a, h\rangle}{\|a\|^3} = 0.$$

The matrix $I - uu^{\mathsf{T}}$ is the orthogonal projection onto the hyperplane perpendicular to a. Thus the derivative removes the radial component and scales the remaining component by $1/\|a\|$.

Applications and further practice

What the approximation says

For a differentiable map,

$$F(a+h) = F(a) + DF(a)h + r(h), \qquad \frac{\|r(h)\|}{\|h\|} \longrightarrow 0.$$

The linear term predicts small changes. A first-order error estimate is not automatically a rigorous bound for a fixed, nonzero measurement error.

8 ⋆ Measurement errors

Use the *measured* values as nominal values. The stated error bounds may be attained simultaneously. Use radians for angles.

(a) **A cylinder.** The volume is $V(r,h) = \pi r^2 h$. The measured radius is $r_0 = 3\,\text{cm}$, with $|\Delta r| \le 0.01\,\text{cm}$, and the measured height is $h_0 = 5\,\text{cm}$, with $|\Delta h| \le 0.005\,\text{cm}$. Use the differential to estimate the largest absolute and relative error in the computed volume. Then find the exact largest absolute error by comparing the endpoint volumes.

(b) **A triangle.** Its area is

$$P(b,c,\alpha) = \tfrac{1}{2}bc\sin\alpha.$$

The measured values $b_0, c_0 > 0$ and $\alpha_0 = \pi/4$ each have relative error at most 1%, for example $|\Delta\alpha| \le 0.01\alpha_0$. Let $P_0 = P(b_0,c_0,\alpha_0)$. Derive an expression for dP/P_0 and show that the *first-order* estimate of the largest relative error is less than 2.8%.

(c) Does part (b) give a strict 2.8% bound on the *exact* relative error $|P - P_0|/P_0$? Find the exact largest relative error over the allowed intervals. *Hint: the area increases with each variable on these intervals.*

Solution

(a) Cylinder. At the measured values,

$$V_0 = 45\pi\ \text{cm}^3, \qquad dV = 2\pi r_0 h_0\,\Delta r + \pi r_0^2\,\Delta h.$$

The largest magnitude of this linear expression is

$$30\pi(0.01) + 9\pi(0.005) = 0.345\pi \approx 1.08385\ \text{cm}^3.$$

The corresponding first-order relative error is

$$2\frac{0.01}{3} + \frac{0.005}{5} = \frac{23}{3000} \approx 0.00766667 = 0.766667\%.$$

Both partial derivatives of V are positive throughout the allowed rectangle. Hence its exact minimum and maximum occur at $(2.99, 4.995)$ and $(3.01, 5.005)$. The exact upward and downward errors from the nominal value are

$$V_{\max} - V_0 = \pi(3.01^2 \cdot 5.005 - 45) = 0.3458005\pi,$$
$$V_0 - V_{\min} = \pi(45 - 2.99^2 \cdot 4.995) = 0.3442005\pi.$$

The exact largest absolute error is therefore $0.3458005\pi \approx 1.08636\ \text{cm}^3$, about 0.768446% of V_0. The linear estimate is close, but is not an upper bound for this finite error.

(b) Triangle: linear estimate. Differentiating and dividing by the nominal area $P_0 = \frac{1}{2}b_0c_0\sin\alpha_0$ gives

$$\frac{dP}{P_0} = \frac{\Delta b}{b_0} + \frac{\Delta c}{c_0} + \cot\alpha_0\,\Delta\alpha.$$

Since $\alpha_0 = \pi/4$ and $\cot\alpha_0 = 1$,

$$\frac{|dP|}{P_0} \le 0.01 + 0.01 + 0.01\frac{\pi}{4} = 0.01\left(2 + \frac{\pi}{4}\right) \approx 0.02785398 = 2.785398\% < 2.8\%.$$

The angular error is 1% of $\pi/4$ radians, not 0.01 radians.

(c) Triangle: exact comparison. The allowed angles lie in $(0, \pi/2)$, where sine is increasing. Therefore

$$\frac{P_{\max}}{P_0} = (1.01)^2\frac{\sin(1.01\pi/4)}{\sin(\pi/4)}, \qquad \frac{P_{\min}}{P_0} = (0.99)^2\frac{\sin(0.99\pi/4)}{\sin(\pi/4)}.$$

The upward relative error is approximately 2.808030%, and the downward relative error is approximately 2.762784%. Thus the exact largest error is about $\boxed{2.808030\%}$. The statement "less than 2.8%" is correct for the first-order estimate, but false as a strict bound on the exact error under the stated assumptions.

9 ⋆ What partial and directional derivatives do not tell us

(a) Define

$$H(x,y) = \begin{cases} \dfrac{x^3}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Prove that H is continuous at the origin and compute every $D_vH(0,0)$ for $v = (p,q) \neq 0$. Find both partial derivatives at the origin. Is H differentiable there? Explain why computing $\nabla H(0,0)^{\mathsf{T}}v$ can give the wrong answer here.

(b) Define

$$K(x,y) = \begin{cases} (x^2+y^2)\sin\left(\dfrac{1}{x^2+y^2}\right), & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Prove from the definition that K is differentiable at the origin. Compute its partial derivatives away from the origin and show that they are not continuous at the origin. What does this say about the condition that the first partial derivatives be continuous near a point?

Solution

(a) Since

$$|H(x,y)| = |x|\frac{x^2}{x^2+y^2} \le |x| \le \sqrt{x^2+y^2}$$

away from the origin, H is continuous at the origin. For $v = (p,q) \neq 0$,

$$H(tp,tq) = t\frac{p^3}{p^2+q^2}, \qquad D_vH(0,0) = \frac{p^3}{p^2+q^2}.$$

In particular, $\frac{\partial H}{\partial x}(0,0) = 1$ and $\frac{\partial H}{\partial y}(0,0) = 0$. Thus its Jacobian $J_H(0,0) = (1,0)$ exists. If H were differentiable, its derivative would necessarily be $DH(0,0)h = h_1$, by the values on the coordinate vectors. But for $v = (1,1)$, the actual directional derivative is $1/2$, whereas this candidate gives 1. Thus H is not differentiable at the origin.

Equivalently, the assignment $v \mapsto D_v H(0,0)$ is not linear:

$$D_{(1,1)}H(0,0) = \tfrac{1}{2} \neq D_{(1,0)}H(0,0) + D_{(0,1)}H(0,0) = 1.$$

One can also see the failure directly in the remainder along $h = (t,t)$:

$$\frac{|H(t,t) - t|}{\|(t,t)\|} = \frac{1}{2\sqrt{2}} \not\longrightarrow 0.$$

Existence of all partial derivatives, even together with continuity and all directional derivatives, does not justify using $D_v H = \nabla H^{\mathsf{T}} v$. Differentiability was an essential hypothesis in Problem 3.

(b) Let $\rho = \sqrt{x^2 + y^2}$. Then $|K(x,y)| \leq \rho^2$. For the zero linear map,

$$\frac{|K(x,y) - K(0,0) - 0|}{\sqrt{x^2 + y^2}} \leq \rho \longrightarrow 0.$$

Thus $DK(0,0) = 0$, and both partial derivatives at the origin are zero. Away from it, write $s = x^2 + y^2 > 0$. Direct differentiation yields

$$\frac{\partial K}{\partial x} = 2x \sin(1/s) - \frac{2x}{s}\cos(1/s), \qquad \frac{\partial K}{\partial y} = 2y \sin(1/s) - \frac{2y}{s}\cos(1/s).$$

Take $x_k = (2\pi k)^{-1/2}$ and $y = 0$. Then $\sin(1/x_k^2) = 0$, $\cos(1/x_k^2) = 1$, and

$$\frac{\partial K}{\partial x}(x_k, 0) = -\frac{2}{x_k} \longrightarrow -\infty.$$

Similarly, $\frac{\partial K}{\partial y}(0, x_k) = -2/x_k$. Neither partial derivative is continuous at the origin. Continuous first partial derivatives in a neighbourhood are a *sufficient* condition for differentiability at a point; this example shows that this condition is not necessary.